# Introduction to Vibe Paper Writing

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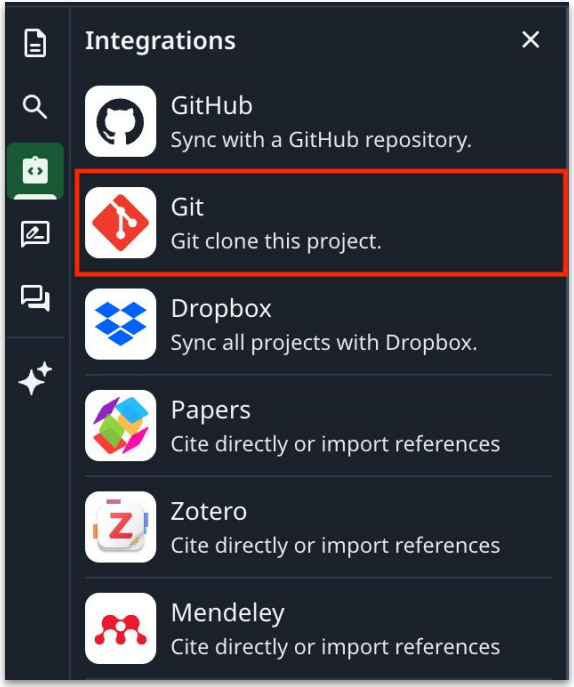
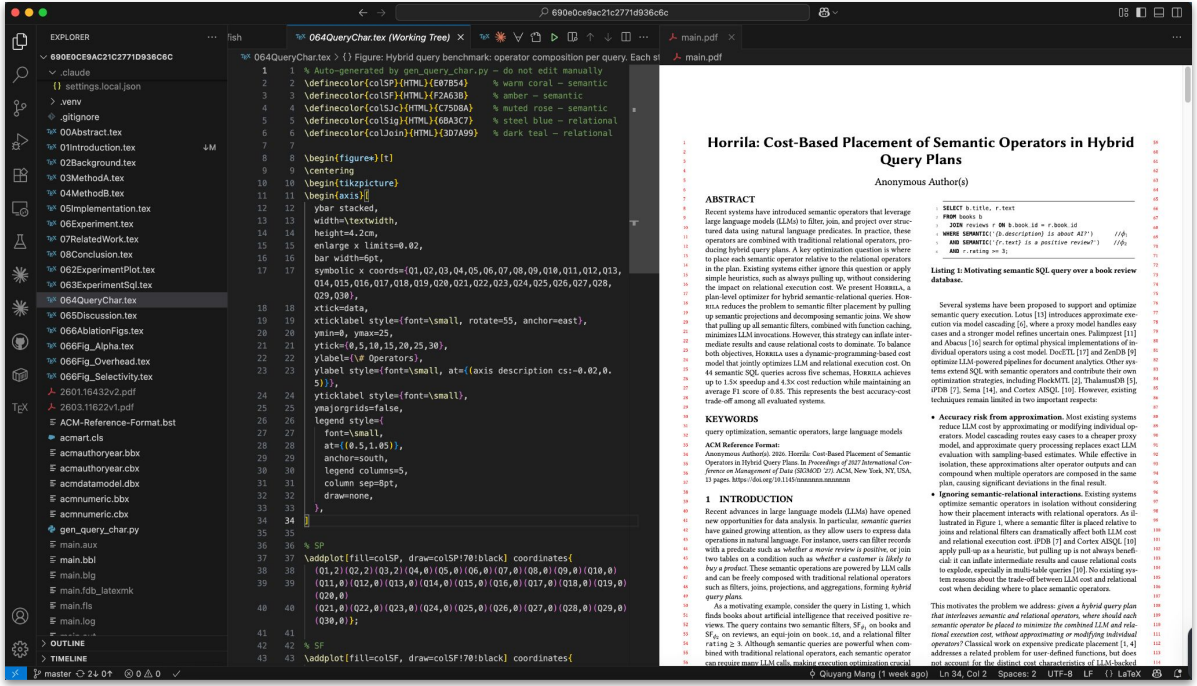
- **Setting Up Your Local AI Writing Environment**
- **AI-Assisted Paper Writing**
  - Polishing Academic Writing
  - Resolving Co-authors’ Comments
  - Drawing Figures
  - AI Review
- **Writing your paper from Scratch with AI**
  - The Top-Down Writing Paradigm
  - Avoiding Hallucinations
  - Using Subagents
  - Minimizing Human-in-the-Loop

# Quick Setup




(1) Set your Overleaf project locally via Git

(2) Writing your paper with local code IDE

- *AI next-line completion*
- *Ask your Agent to make edition go through the project*

# Polishing your writing

What's or

You're a professional research paper ...
Polish the following sentences in my research paper [sentences]

[context]

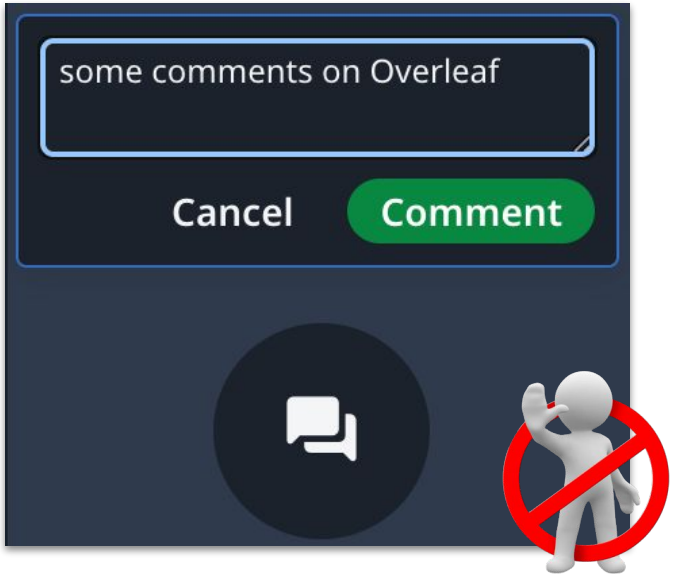
+

```
407   No newline at end of file

● Update(064QueryChar.tex)
  └ Added 1 line, removed 1 line
    68  \end{axis}
    69  \end{tikzpicture}
    70  \caption{Hybrid query benchmark: operator composition per query. Each stacked bar shows the count of $
        \SP$, $\SF$, $\SJ$, $\sigma$, and $\Join$ operators. Q17--Q19 and Q26--Q30 are the most complex, with
        up to 9 joins and 4 semantic filters.}
    71 -\Description{Stacked bar chart showing operator counts per query across all 30 benchmark queries.}
    71 +\Description{Stacked bar chart showing operator counts per query across the 30-query hybrid benchmark.
       +}
    72  \label{fig:query-char}
    73  \end{figure*}

● Bash(pdflatex -interaction=nonstopmode -halt-on-error main.tex 2>&1 | grep -E "^!|Fatal|Output written")
  └ Output written on main.pdf (13 pages, 662691 bytes).

● Done. All changes:

  \sky{} removed:
  - 06Experiment.tex:83 → "Hybrid Query Benchmark Results"
  - 06Experiment.tex:131 → "SemBench E-Commerce Results"

  30 → 44 updated in:
  - 00Abstract.tex — "44 semantic SQL queries across five schemas"
  - 08Conclusion.tex — "44 queries across five schemas"

  30-query references clarified as "30-query hybrid benchmark" in:
  - 06Experiment.tex — table caption, metrics paragraph, results text
  - 062ExperimentPlot.tex — figure caption and description
  - 064QueryChar.tex — description

  The 30-query hybrid benchmark + 14 SemBench queries = 44 total, consistent throughout.

✻ Worked for 3m 8s

>
```

Bad practice: Copy-pasting your writing in

Ask your claude code (with better context)

# Resolving Co-authors' Comments




Bad practice: Leaving comments on Overleaf interface

```
To do so, we ask the generator to produce additional columns and tables in the schema, and then use
those \emph{virtual columns} and \emph{virtual tables} to represent the two types of
\modifyb{placeholders}, respectively, \emph{i.e.,} $\mathsf{Expression}$ and $\mathsf{Table}$ shown in
\Cref{fig:bnf-grammar}.
% \alvin{shown in Fig XXX}
As shown in Listing~\ref{lst:virtual-column-and-table}, we create a virtual column
\texttt{placeholder1} to represent an expression, \emph{i.e.,} $\square_1
\triangleright\placeholder{\textsf{Expr}(t1:BOOLEAN)}$, and a virtual table \texttt{vtable1} for a
table placeholder \emph{i.e.,} $\square_1 \triangleright\placeholder{\textsf{Table}(t0:INT, t1:INT)}$.
% \alvin{why are they crossed off? looks weird}
The generator, unaware of this abstraction, directly uses the concrete \texttt{placeholder1} and
\texttt{vtable1}, effectively producing a valid CAQ.
This approach makes our CAQ representation compatible with SQL provers expecting concrete syntax and
easy for LLMs to understand.
% \alvin{need a better way to explain this. let's discuss}
```

Direct leaving your comments in *.tex*

```
❯ find all comments by \alvin{} & solve them (for each one, analyze the context, and propose several candidate patch
  let me decide finally)
```

# Drawing Figures

**Bad Practice**: AI-generated Python Code then place into the paper

**Good Practice**: TIKZ + Claude Code is all your need!

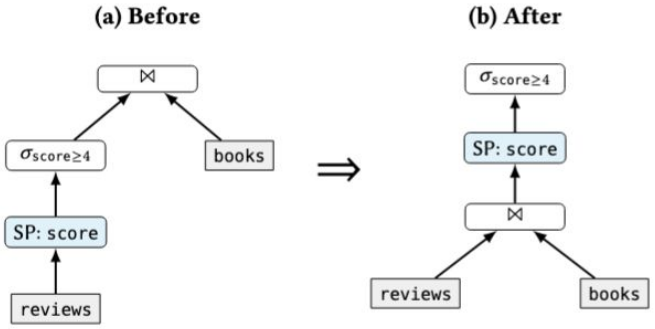
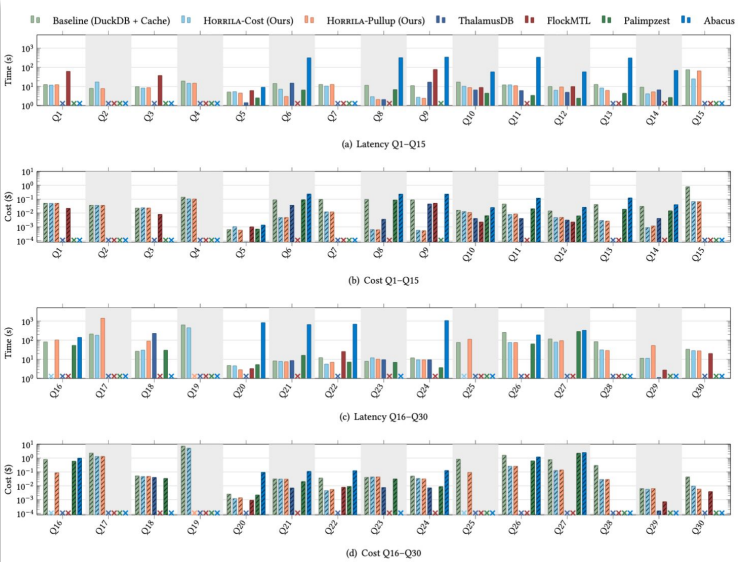
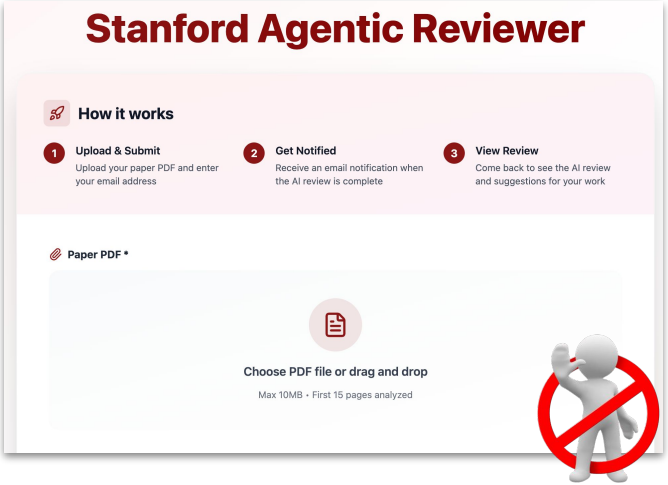



Figure 2: Pulling up SP and its dependent $\sigma$. (a) SP computes a sentiment score on all reviews before the join. (b) After pull-up, SP evaluates only reviews that have a matching book.



# AI Paper Review




Bad practice:
Using AI paper review website

```
❯ Build a sub review agent based on this https://paperreview.ai/tech-overview

● Fetch(https://paperreview.ai/tech-overview)
  └ Received 18.7KB (200 OK)

● Now let me build the agent with this pipeline.

● Wrote 1 memory (ctrl+o to expand)

● Now update the memory index and launch the agent.

● Recalled 1 memory, wrote 1 memory (ctrl+o to expand)

● Now launch the agent.
```

Build your local review subagent

- *Do self-play with your writing agent*

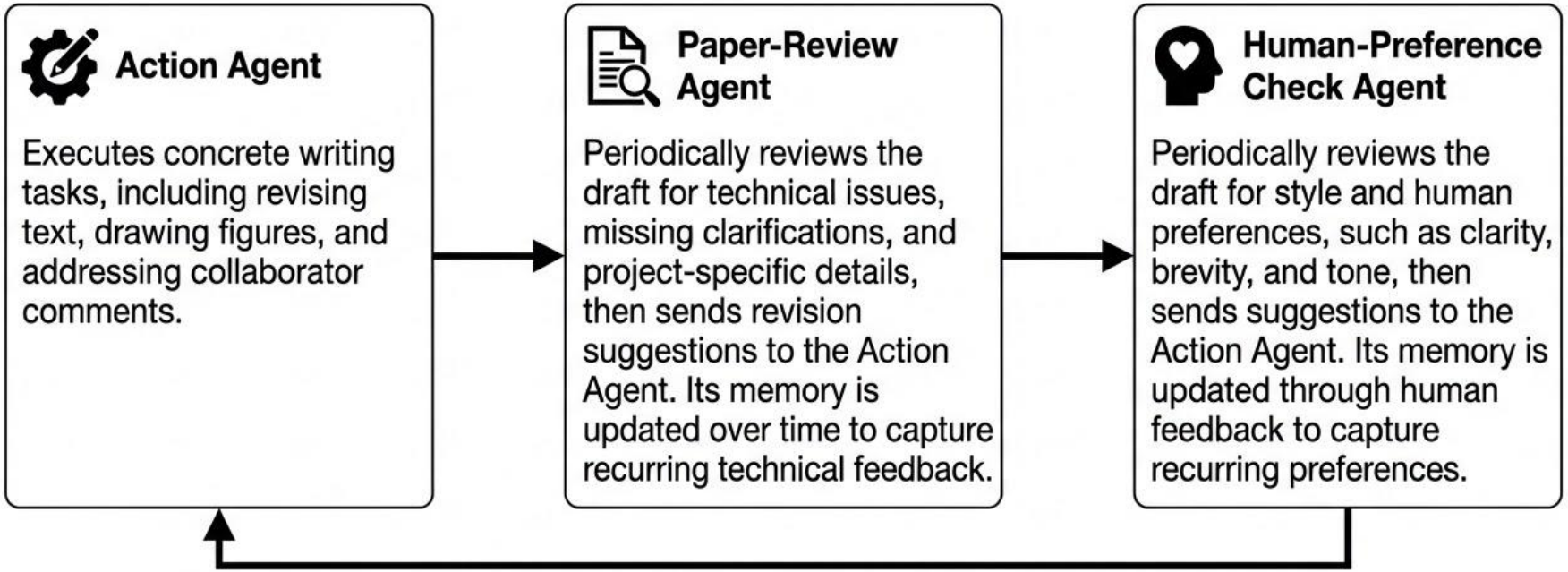
# AI writing from scratch

1. Place a few reference papers in the same directory—ideally from the same target conference but different research areas (to avoid plagiarism).

2. Traditional writing usually **starts from concrete sections** like methods and experiments, and only later moves to the introduction, abstract, and conclusion. Vibe writing flips this workflow: **start with a few high-level bullets (abs/intro)**, then expand them top-down into a full paper.

3. Humans still need to stay in the loop to ensure that the AI generates accurate content and fully understands the details of the project.

# AI writing from scratch with subagents



# Thanks!

Fake or Real paper? Try it out!